

History and Applications of Molex Connectors

Abstract

Molex connectors have played a pivotal role in the electronics industry for over half a century. This paper presents a comprehensive overview of Molex – from the company’s origins in 1938 through the development of its connector technologies – and examines the technical specifications and evolution of these connectors over time. We discuss the widespread applications of Molex connectors across domains such as robotics, consumer electronics, and the automotive industry, highlighting how their reliability and standardization contributed to ubiquity in these fields. We also analyze why certain Molex connector types (notably the four-pin power plug in PCs) have attained *legacy* status, having been superseded by newer standards in some contexts. Diagrams and figures are included to illustrate common Molex connector types and their internal structures. Through an expert lens, this paper underscores the historical significance of Molex’s innovations in connectivity, and how its connectors continue to impact modern engineering despite the emergence of new technologies. All information is cited from reputable sources, and references are provided in IEEE format to ensure accuracy and traceability.

Introduction

Molex Incorporated is one of the world’s leading manufacturers of electronic connectors, with a product portfolio of over 100,000 interconnect products serving industries from data communications and consumer electronics to automotive and medical devices ¹ ². Founded in 1938 in Illinois, USA, the company has grown over 80+ years into a global supplier and is consistently ranked among the top connector companies by annual revenue ¹. Molex is especially renowned for pioneering the **Molex connector** – a two-piece pin-and-socket plug introduced in the mid-20th century – which achieved universal adoption in personal computers and other electronics ³. Notably, the term “*Molex connector*” in common usage often refers specifically to a 4-pin 5.25” drive power connector used in PCs, though Molex as a company produces a vast range of connector types ⁴ ⁵. This paper will clarify this terminology and explore the breadth of Molex’s connector innovations.

We begin with a historical background of the Molex company and the development of its connector technology. Next, a technical overview is provided, explaining the specifications of major Molex connector families and how they evolved over time. We then delve into key application domains – including robotics, consumer electronics, and automotive – to illustrate how Molex connectors are utilized in each, and how they became standard solutions in those arenas. In a subsequent section on *Legacy Status*, we discuss certain Molex connectors that have become legacy interfaces in modern designs, examining the reasons behind their decline or replacement in specific contexts. Finally, we conclude with a summary of Molex’s enduring impact on connector technology. Throughout, an expert tone is maintained and claims are supported with citations from datasheets, industry publications, and historical records.

Historical Background

Molex was established in 1938 by **Frederick August Krehbiel** as the Molex Products Company, originally not as an electronics firm but as a manufacturer of plastic molding compounds and consumer items ⁶ ⁷. The company's very name "Molex" comes from a proprietary plastic material developed by Krehbiel in the 1930s – a composite of coal tar pitch, limestone, and asbestos fibers – which was found to be an excellent electrical insulator ⁸. In its early years, Molex used this material to produce household products like flower pots, salt shakers, and clock cases ⁶. By the 1940s, as World War II drove demand for plastics, Molex's insulating material gained attention because it was not subject to wartime rationing (unlike many standard plastics) ⁹. This positioned the company advantageously to explore new product lines in the electrical domain.

Post-war, shifting to connectors: After WWII, under the leadership of Frederick's son **John H. Krehbiel**, Molex identified a growing market for electrical connectors. The company shifted focus from just molding materials to designing interconnection devices, marking a pivotal transition from a materials company to a connector company ¹⁰ ¹¹. In 1953, Molex launched its first plug-and-receptacle connector line, targeting the burgeoning consumer appliance industry ¹². These early connectors were low-cost terminal blocks and simple pin-and-socket connectors that found success in home appliances such as refrigerators and washing machines ¹². By 1960, Molex had introduced its first connectors made of nylon (rather than the original Molex plastic), signifying its full evolution into an electronics connector manufacturer ¹³ ¹¹. This development of nylon plug and receptacle lines in 1960 was critical – nylon's durability and heat resistance made it ideal for the increasingly complex electrical applications of the era.

Rise of the Molex connector in electronics: Molex's breakthrough came in the late 1950s and 1960s by supplying connectors to the fast-growing consumer electronics market. Notably, Molex developed pin-and-socket connector designs for use in **color televisions**, which were then an emerging technology ¹⁴. These connectors, designed by John Krehbiel, linked electronic components inside TV sets and were adopted by major TV manufacturers like Zenith, RCA, and Magnavox ¹⁵. **Figure 1** provides an example of a Molex standard pin-and-socket connector assembly from that period, showing the male and female plastic housings alongside the removable crimp contact pins.

Figure 1: Components of a classic Molex pin-and-socket connector (Disk Drive Power Connection System). Shown here are a 4-pin Molex connector male housing (plug) and female housing (receptacle), corresponding crimp terminals (male pin and female socket), and the specialized hand crimp tool used for assembly ¹⁶. The pin-and-socket design, first patented by Molex in the late 1950s, features cylindrical metal pins that insert into spring-metal sockets, all housed in a nylon shell ¹⁷. This basic design became a de facto standard for power connectors in many electronics.

By 1962, the demand for Molex's connectors had grown the business substantially – the company's annual sales exceeded \$1 million, driven largely by the widespread use of its connector products in consumer electronics ¹⁴. Molex earned a reputation for solving challenging design problems through unique interconnect solutions ¹⁸. For example, the company developed **simple PCB connectors and stamped circuitry**, as well as the aforementioned nylon plugs and receptacles for color TVs ¹⁸. In the 1960s and 70s, Molex also introduced progressively **smaller connector form factors**, essentially the "micro" connectors of their day, to keep pace with miniaturization trends in electronics ¹⁸. This early emphasis on innovation laid the groundwork for Molex to become a leader in connector miniaturization – a trend that continues in its product development to this day.

Global expansion and growth: As its connector business expanded, Molex grew from a small Illinois enterprise into a global company. It opened its first overseas factory in **Yamato City, Japan in 1970**, allowing it to better serve the rapidly growing Japanese electronics industry ¹⁹. In 1972 Molex went public, using the infusion of capital to accelerate growth in international markets ²⁰. Through the 1970s and 1980s, Molex further broadened its product lines (entering markets like automotive and computer peripherals) and aggressively expanded manufacturing globally – by the early 1990s Molex operated more than 40 plants in 19 countries ²¹. This global footprint enabled close collaboration with diverse customers and contributed to Molex's emergence as (by 1993) the second-largest connector company worldwide (trailing only AMP Incorporated) ²¹. Key acquisitions also enhanced its capabilities; for instance, in 1980 Molex acquired Cardell Corporation, a maker of automotive connectors, strengthening Molex's position in the automotive connector market ²² ²³.

21st century developments: In the 2000s, Molex continued to thrive by focusing on high-performance connector solutions and niche innovations. The company invested heavily in R&D (routinely over 5% of annual revenue) to develop new connector families for emerging needs such as high-speed data communication and mobile devices ²⁴. By 2013, Molex had over 45,000 employees and \$3.6 billion in annual revenue ²⁵ ²⁶. That year, the Koch Industries conglomerate acquired Molex, though Molex retained its name and focus on connectors. Today, after more than 80 years, Molex stands as a cornerstone in the connector industry. It has built “a legacy of innovation” from its humble start with molded plastics to modern high-density, high-speed connectors ²⁷. The historical trajectory of Molex is marked by continuous adaptation to new technologies and markets, which is reflected in the wide array of connector types bearing the Molex name.

Technical Overview of Molex Connectors

Molex connectors encompass a vast range of interconnect types, but many share common design principles and have evolved along a few fundamental families. At their core, most Molex connectors are **two-piece, pin-and-socket** systems: one half contains one or more metal pin contacts (often on the male or plug side) and the mating half contains corresponding socket contacts (on the female or receptacle side). These contacts are typically housed in molded plastic (originally cellulose-based plastic or nylon) shells that provide insulation, alignment, and locking features ¹⁷. This section outlines the technical characteristics and evolution of Molex connectors, from the classic power plugs to modern miniaturized connectors.

Pin-and-socket design: The archetypal Molex connector, as developed in the late 1950s, features **cylindrical spring-metal pins** that fit into matching spring-metal sockets ¹⁷. Both pins and sockets are held in a rectangular insulating housing, and the connectors are usually **polarized/keyed** to prevent incorrect orientation ²⁸. Early examples like the Molex **0.093-inch series** used relatively large diameter pins (~2.36 mm) capable of carrying about 8.5 A each ²⁹. These became popular for delivering power in electronics due to their robust current capacity and simple crimp termination. Over time, smaller pin sizes (such as 0.062” or 0.084” diameter) were introduced for lower-current or higher-density needs, with the smallest in this family handling around 5 A per pin ²⁹. **Mechanical latching** mechanisms were also implemented: while some early connectors relied purely on friction fit to hold the halves together, later designs added latch tabs or locking ramps to ensure a secure connection that would not accidentally unplug ³⁰. For instance, many Molex housings have integrally molded snap latches or require an audible “click” on mating, confirming proper engagement.

Despite widespread adoption, the classic friction-fit Molex connectors did have known issues. They could be **difficult to disconnect** due to tight fit and lack of a release lever, often requiring significant force to unplug ³¹. Additionally, if the female socket contacts lost tension or the male pins were misaligned, the connection could loosen over time, leading to contact resistance and **electrical arcing** (visible as browning or melting of the nylon housing in severe cases) ³². These issues spurred continuing refinement of connector designs by Molex and others, focusing on better locking features and contact stability in later generations.

Common Molex connector families: As electronics diversified, Molex developed numerous connector series tailored to different pitches (pin spacings), current ratings, and environments. Some of the most significant families include:

- **KK™ series (Modular Connectors):** The KK series is a family of **low-power board-to-board and wire-to-board connectors** with a typical pitch of 2.54 mm (0.100") ³³. Introduced in the 1970s, Molex KK connectors are often seen as simple plastic housings that accept crimped tin-plated terminals; they became a “building block” system that could be configured in various pin counts. These are widely used for signal connections on PCBs, in everything from home appliances to hobbyist electronics. The popularity of KK connectors led to many compatible clones – for example, the ubiquitous 2.54 mm pitch plug and socket pairs found in PC motherboard front-panel headers and Arduino-style kits are often Molex KK or compatible designs ³³ ³⁴. The KK family has variants with friction ramps or latching features to secure the connection ³⁵. Its enduring presence (despite newer alternatives) is due to the simplicity and standard 0.1” spacing that matches PCB grid layouts.
- **Mini-Fit Jr.® series:** This is a family of **medium-power connectors** with a pitch of 4.20 mm, featuring fully isolated rectangular pins and positive locking via latch tabs. Molex’s Mini-Fit Jr., introduced in the 1990s, is notable for being the connector system used in ATX PC power supply cables (for example, the 20/24-pin main motherboard power connector is a 24-pin Mini-Fit Jr. plug) ³⁶. Each Mini-Fit Jr. contact can carry up to about 9 A with 16 AWG wire ³⁷ ³⁸, making it suitable for delivering substantial power in a relatively compact form factor. The housings are polarized and have locking clips to prevent accidental disconnection. In addition to PC power connectors, Mini-Fit Jr. and its higher-power cousin Mini-Fit Sr. find use in industrial and automotive systems that require rugged power connectors. The widespread adoption of Mini-Fit Jr. in computer hardware (both by Molex and equivalent connectors from competitors) underscores how a Molex design became an *industry standard* for modular power delivery ⁵ ³⁹.
- **Disk Drive Power connectors (Molex 8981 series):** Among Molex’s most famous products is the 4-pin power plug used historically for computer disk drives and peripherals (often just called “the Molex connector” by PC enthusiasts). This connector, originating in the late 1970s, has a 5.08 mm pitch with 0.093” diameter contacts providing 5 V and 12 V DC power and ground ⁴⁰. Interestingly, the first versions of this connector used in PCs were based on the **AMP/Tyco Mate-N-Lok** series introduced in 1963 ⁴¹ ⁴². Molex later produced a compatible version (Series 8981, trademarked as the Disk Drive Power Connection System) in 1983 ⁴³. The standard configuration is a 4-pin arrangement in a row, with a friction fit (no latch) – a design that proved very reliable for decades of computer hardware ⁴⁴. **Figure 2** shows a close-up of this connector type in use, along with the crimp terminals that are inserted into the housing. While robust, this connector lacks modern features like hot-swap capability and has largely been phased out of new PC designs (see Legacy Status section), yet it remains a hallmark of Molex’s influence on PC architecture.

Figure 2: Example of a classic 4-pin Molex peripheral power connector (two-piece pin and socket). The image shows a mated connector providing power inside a device, with loose female crimp pins (on the right) that would be inserted into the plug housing and male pins in the receptacle on the device side. This connector delivers +5 V (red wire) and +12 V (yellow wire) with two ground wires (black) ⁴⁵. Developed in the 1960s, the 4-pin Molex power connector became a standard for disk drives and other peripherals through the 1980s and 1990s ⁴³.

- **Micro-Fit 3.0™ series:** As the need arose for connectors smaller than Mini-Fit but still capable of moderate power (up to ~5 A per circuit), Molex introduced the Micro-Fit family with a 3.00 mm pitch. Micro-Fit connectors are high-density, dual-row connectors commonly used in compact PCs, robotics, and industrial control for both power and signal. They include features like positive locks and keyed housings to avoid mis-mating. For instance, in robotics applications, teams have used Micro-Fit 3.0 connectors for sensor and actuator wiring because of their balance of size and current capacity ⁴⁶. However, as noted by practitioners, proper crimping and seating of the tiny terminals can be challenging, prompting exploration of alternatives for reliability ⁴⁶.
- **Microminiature and Specialty connectors:** Molex has continually pushed toward smaller connectors to serve modern electronics. Families like **PicoBlade™ (1.25 mm pitch)** ⁴⁷ and **NanoBlade™/MicroBlade** are used for low-current signal connections in consumer gadgets, laptops, and wearables, where space is at a premium. These often employ a friction latch mechanism and a two-point contact design to ensure reliability even at miniature scale ⁴⁷. In high-vibration or harsh environments, Molex also offers *sealed connectors* (e.g., ML-X™, MX150™) with rubber grommets and robust housings, and *RF/coaxial connectors* for high-frequency signals. The **MX150** family cited earlier is a sealed automotive connector system with a 1.50 mm blade-type interface, supporting up to 22 A and temperatures of 125–150 °C for under-hood applications ⁴⁸. Another modern example is Molex's **SlimStack®** board-to-board connectors, which have fine pitches down to 0.35 mm for connecting printed circuit boards in devices like smartphones; despite their tiny size, they include alignment features and even audible click feedback for mating ⁴⁹.

Throughout its connector lines, Molex has emphasized compatibility and incremental improvement. Many of its newer connectors maintain backward-compatible footprints or mating interfaces with older ones (when possible) to ease adoption. The company also provides extensive **application tooling** – from hand crimpers (as in Figure 1) to automated press-fit machines – reflecting the importance of proper termination in connector performance. By offering both the connectors and the tools to apply them, Molex helped ensure that its connectors could be reliably integrated by manufacturers at scale ⁵⁰.

In summary, the technical evolution of Molex connectors has been characterized by: (1) a **reduction in size (pitch and form factor)** without sacrificing mechanical and electrical integrity, (2) improvements in **current density** (more current per pin or per unit area through better materials and designs), (3) enhanced **mechanical latching and keying** features to prevent errors and accidental disconnections, and (4) specialized variants for **harsh environments** (sealed, high-temperature connectors for automotive and industrial use). These innovations have enabled Molex to supply interconnect solutions from the largest power connectors down to the tiniest microminiature links, cementing the company's role as a one-stop interconnect provider.

Application Domains of Molex Connectors

Molex connectors are employed across an exceptionally broad range of industries. Here we detail their usage and significance in three key domains – **robotics**, **consumer electronics (including computing)**, and **automotive** – while noting that they are also prevalent in telecommunications, medical devices, aerospace, and more. Each domain often utilizes specific Molex connector series optimized for that environment's requirements.

Robotics

In robotics (encompassing industrial automation robots, mobile robots, and even competitive robotics teams), reliability and robustness of electrical connections are critical. Robots typically have numerous actuators (motors, servos), sensors, and control boards interconnected by cables – all of which must withstand motion, vibration, and sometimes harsh conditions. Molex connectors are a popular choice in robotics for both **power and signal wiring** due to their proven performance.

For power distribution in robots, teams frequently use Molex's high-current connector families. For example, the **Mini-Fit Sr.**TM series (the larger cousin of Mini-Fit Jr.) is often chosen for battery and motor controller connections because it can handle heavy gauge wires (12 AWG or larger) and currents well above 20 A with secure locking ⁴⁶. One FIRST Robotics Competition team noted using Mini-Fit Sr. for all 12 AWG power runs on their robot ⁵¹. In the same context, smaller Molex connectors like Micro-Fit 3.0 have been used for lower-current sensors and devices ⁴⁶. These connectors enable modular wiring harnesses – sections of the robot can be quickly connected or replaced via plug-in cables rather than soldered joints, which is advantageous for maintenance and design flexibility.

Molex also provides connector solutions specifically targeting robotics and automation. The **Molex Industrial Automation** product line includes heavy-duty circular connectors, M12 sensor connectors, and the Compact Robotic Connectors (CRC) series for slim actuators ⁵² ⁵³. In educational robotics (like FIRST), Molex has introduced an **SLTM series** kit – containing standardized **SL connectors** for PWM signal cables, CAN bus connections, etc., along with pre-crimped leads for teams without professional crimp tools ⁵⁴ ⁵⁵. The SL series is a simple wire-to-wire and wire-to-board connector family with locking housings, well-suited for signal wires (often 22–30 AWG) on robot control systems. By providing such kits, Molex has essentially helped standardize wiring practices in the educational robotics community, improving reliability (e.g., reducing loose or intermittent connections from improper wiring) ⁵⁵.

In industrial robotics, connectors must endure continuous motion (robot arms flexing cables) and often exposure to oil or dust. Molex addresses these with **flexible cable connectors and sealed interfaces**. For instance, the **Brad® line** (originating from Molex's acquisition of Woodhead) offers rugged circular connectors and quick-disconnect systems for industrial sensors and actuators. Additionally, Molex's expertise in miniaturization is now aiding *consumer robotics*: small home robots or drones benefit from micro connectors like PicoBlade that save space and weight while delivering reliable connections ⁵⁶. As robots become more sophisticated, high-speed data connectors (for machine vision cameras, LiDAR, etc.) are also needed; Molex supplies solutions here too (e.g., automotive-grade ethernet connectors used in autonomous robots). In sum, Molex connectors in robotics range from tiny board-level connectors on controllers to high-power battery plugs – all designed to maintain connectivity under the dynamic conditions unique to robotics.

Consumer Electronics and Computing

Molex connectors have long been ubiquitous in consumer electronics – a domain that includes personal computing devices, home entertainment equipment, appliances, mobile devices, and more. Their widespread use in this area is due to a combination of standardization, cost-effectiveness, and the vast array of form factors Molex offers for different needs.

In the realm of **personal computers**, Molex connectors essentially set the standard for internal power and board connections for decades. Early microcomputers and PCs in the 1970s–90s employed Molex pin-and-socket connectors for power supply interfaces: for example, IBM PCs used two 6-pin Molex connectors (the P8/P9 connectors) to link the PSU to the motherboard in the 1980s ⁵⁷. By the mid-1990s, the ATX standard introduced the 20-pin (later 24-pin) main power connector, which – as mentioned – is a Molex Mini-Fit Jr. series plug ³⁶ ⁵⁸. Molex's connectors were also used for peripheral devices: the 4-pin **Molex peripheral power connector** (Figure 2) became a **de facto standard for hard drives, optical drives, and other peripherals** across all PC manufacturers ⁴³ ⁵⁹. Additionally, smaller **Berg connectors** (another brand, but often supplied by Molex) were used for floppy drives' power. Inside a typical desktop PC from around 2000, one would find a dozen Molex-made or Molex-compatible connectors: motherboard headers for USB, front panel LEDs (often KK family), fan power connectors, drive power connectors, and so on.

Molex connectors gained this universal adoption in PCs in part because Molex actively worked with standards committees (like the Intel ATX specification) to define connector interfaces. For instance, Intel's ATX specification explicitly references Molex part numbers for the main power connector and peripheral power connectors as approved reference designs ⁵. The consistency and reliability of these connectors helped ensure interoperability – any ATX power supply could plug into any ATX motherboard because they all used the same Molex Mini-Fit Jr. 20/24-pin interface. This standardization was crucial for the growth of the PC industry, enabling mix-and-match of components from different vendors.

Beyond computers, in **consumer electronics and appliances**, Molex connectors appear virtually everywhere behind the scenes. Televisions and audio equipment often use wire-to-board KK or MicroBlade connectors to link power boards, speakers, and control boards. Household appliances (washing machines, HVAC systems, kitchen appliances) use a variety of Molex quick-connect harnesses – e.g., **Molex .093" series or MLX series** connectors for internal wiring harnesses that carry power to motors and sensors. These connectors offer secure latching and the ability to carry higher currents for things like compressor motors. The Molex Mini-Fit Jr. series, while famous in PCs, is also found in appliances that need compact connectors for moderate power – for example, some **refrigerators and air conditioners** use 2- to 4-circuit Mini-Fit connectors to hook up control boards and fans, taking advantage of their 6–9 A per pin capability and reliable locking ⁶⁰. In high-volume consumer products, cost is a major factor; Molex often provides value-engineered connector versions (using cheaper materials or simpler designs) to meet price targets while maintaining adequate performance.

An area of increasing importance is **mobile and handheld devices**. As gadgets like smartphones and tablets became thinner, traditional wire harness connectors were too bulky, which spurred development of micro coaxial connectors, board-to-board mezzanine connectors, and flex cable connectors. Molex has been at the forefront here, creating families like **SlimStack board-to-board connectors** used to connect display panels or modules inside phones ⁴⁹, and **Micro USB/USB-C connectors** (Molex is a supplier of standardized USB connectors for charging and data). For wearables and IoT devices, Molex micro connectors (like *micro coax RF connectors* and tiny wire-to-board connectors) enable sensors and batteries to

be connected in extremely tight spaces ⁶¹. The company's focus on miniaturization (noted in the historical section) shows in these products, some of which are nearly invisible yet deliver high-speed signals or significant power.

In summary, Molex connectors in consumer electronics are prevalent because they offer **standard solutions that manufacturers trust**. Whether it is the backplane connector in a plasma TV, the battery connector in a smartphone, or the pin header in a home alarm system, using a Molex connector often means leveraging decades of refinement. This translates to fewer connection failures in the field and readily available mating pairs and crimp tools worldwide. The phrase “when the word Molex is used by the computer enthusiast it refers to the standard power connectors found in modern computers” ⁶² succinctly captures how ingrained Molex's connectors became in consumer tech vernacular.

Automotive

The automotive industry presents one of the most demanding environments for connectors: extreme temperatures, moisture, vibration, and safety-critical reliability requirements. Automobiles contain dozens of electronic control units (ECUs) and hundreds of sensors/actuators, all linked by wiring harnesses. Molex has been deeply involved in automotive connectors for many decades, both through its own developments and strategic acquisitions (such as Cardell in 1980, and more recently partnerships in autonomous vehicle connectors).

Traditionally, automotive wiring harnesses used a variety of specialized connector systems (often following standards like USCAR). Molex's early involvement in automotive included supplying connectors for basic electrical systems and later for the first generations of electronic engine control and infotainment. A hallmark of Molex's automotive connectors is the focus on **sealed connectors** to prevent ingress of water, dirt, and chemicals. For example, **Molex MX150™** is a family of *sealed* and *unsealed* connectors widely used in car engine compartments and lighting systems ⁴⁸. The MX150 series conforms to USCAR specifications and uses a mat-seal design (integral seals) to achieve environmental sealing. It can operate up to 150 °C and carry up to 22 A per circuit, making it suitable for applications like fuel injectors or exterior lighting ⁴⁸. Despite being a mature product, MX150 remains a go-to solution in automotive due to its proven field performance and standardized interface (it provides roughly 37% space savings over older 1.50 mm pitch connectors without sacrificing current handling) ⁶³.

As vehicles demanded more electronic features, the number of connections skyrocketed, leading to a push for *smaller automotive connectors* to save space and weight. Molex answered this with the **Mini50™** connector system – the industry's first automotive-grade connector with a 0.50 mm terminal interface (roughly 50% smaller footprint than the traditional 0.64 mm connectors used in many car harnesses) ⁶⁴. Mini50 connectors are designed for low-current signals in space-constrained locations, such as in *infotainment systems, advanced driver-assistance systems (ADAS) sensors, and body electronics*. They are available in both sealed and unsealed versions and meet USCAR requirements, illustrating how Molex brought miniaturization to the automotive world without compromising on ruggedness ⁶⁴. By reducing size, car manufacturers can route harnesses through tighter spaces and potentially reduce the weight of bundles by including more circuits in a smaller connector housing.

Additionally, Molex has developed high-speed data connectors for automotive applications. Modern cars use Ethernet-based networks, high-speed camera links, etc. Molex's connectors like **HSAutoLink™** and coaxial RF connectors help implement USB, LVDS camera links, and Ethernet in cars with automotive-grade

robustness. For example, the HSAutoLink family supports data connectivity (for infotainment and telematics) and leverages Molex's background in telecommunications connectors, ruggedized for vehicles. As vehicles move towards electrification, high-voltage connectors are also needed (for battery packs, power electronics). Molex (along with other major connector companies) produces high-voltage cable connectors and busbar connectors that adhere to automotive safety standards for hybrid and electric vehicles.

One cannot overlook the importance of **legacy support in automotive**: Car models typically remain in service for decades, and spare parts must be available. As such, the industry tends to stick with proven connector designs for long periods. Molex's legacy connectors (like the older MLX series or CGrid and KK series for interior electronics) still appear in new vehicles for this reason. Manufacturers appreciate that a mature connector series will have stable supply and known failure rates ⁶⁵ ⁶⁶. Moreover, reusing an existing connector design means technicians are already trained to assemble and repair them, and OEMs have already invested in the crimp tooling and assembly equipment ⁶⁶. Molex, recognizing this, continues to produce and support many of its legacy automotive connectors even as it introduces new ones – balancing innovation with backward compatibility.

In summary, Molex's role in automotive connectors is characterized by a dual focus: **continuous innovation** (smaller, smarter connectors for next-gen vehicles) and **sustaining reliable legacy systems** (ensuring that existing connector styles remain available and improved incrementally). This strategy helps automakers integrate new features without reinventing the wheel for every connector interface, contributing to vehicle reliability and manufacturability.

Legacy Status of Molex Connectors

Certain Molex connectors, despite their historical prevalence, have become regarded as “legacy” in modern contexts – meaning they are largely phased out or used only for backwards compatibility with older equipment. This section explores why this occurs, using the example of the **4-pin Molex peripheral power connector** in PCs as a primary case study, and touches on other scenarios where older Molex connector styles are superseded by newer technology.

The 4-pin Molex power plug in PCs: For decades, the white 4-pin Molex plug (Figure 2) was the standard way to supply disk drives and fans with 5 V and 12 V from a PC power supply. However, with the advent of the Serial ATA (SATA) interface in the mid-2000s, a new power connector was introduced for hard drives and optical drives. The SATA power connector is a 15-pin wafer-style connector that provides additional voltages (including 3.3 V) and supports hot-swapping drives ⁵⁹. This new design, which Molex and other companies helped develop, offers a more streamlined form factor and features like staggered pins for hot-plugging. As SATA drives became ubiquitous, the older 4-pin Molex plug saw diminishing use. New drives stopped including Molex power sockets, favoring the SATA power port exclusively. Consequently, in a modern PC, the 4-pin Molex connectors from the power supply might only be used for peripherals like case fans, water cooling pumps, or LED lighting strips – and even many of those have transitioned to newer connectors or direct motherboard connections.

Today, the 4-pin Molex is widely considered a *legacy connector* in the PC world ⁶⁷. Enthusiast forums commonly describe it as “really old” and note that “barely anything still uses Molex” in a contemporary computer build ⁶⁷. Its design limitations (no latch, stiff to unplug, bulky shape) and the lack of a 3.3 V supply made it less suitable for newer devices. The SATA power connector's slim, keyed design is easier to insert and cannot be accidentally reversed, improving user experience and safety ⁶⁸. Moreover, the SATA

connector's additional pins allowed for future features (e.g., device power management signals) that the simplistic 4-pin Molex could not support.

However, PC power supply manufacturers have continued to include a few Molex outputs on most units, primarily for compatibility with older accessories. For instance, even a modern mid-range PSU might come with two or three 4-pin Molex plugs and perhaps an adapter for a floppy drive (legacy upon legacy!) ⁶⁹. This is because some users may still need to power an older DVD drive, a fan controller, or other hardware that hasn't updated to SATA power or newer connectors. Adapters are also readily available (e.g., Molex-to-SATA power adapters) to bridge old and new: a Molex plug can be repurposed to supply a SATA drive via a simple adapter cable ⁷⁰. This backwards compatibility ensures that the transition away from Molex is gradual and does not render existing equipment unusable. But effectively, by the late 2010s, the Molex 4-pin power connector joined the ranks of ISA slots and parallel ports as relics of PC history – important to remember, but rarely seen in cutting-edge systems.

Other legacy contexts: Beyond the PC power plug, there are other Molex connectors that have been supplanted by new solutions in various fields:

- In **electronics hobbyist and small devices**, the older 2.54 mm KK-series connectors or larger pitch headers are sometimes viewed as legacy as the industry shifts to smaller JST-brand connectors (e.g., JST-PH 2.0 mm or JST-SM series) for compactness. While Molex KK connectors are still used (especially in prototyping and larger devices), many modern consumer gadgets have no 0.1" connectors internally, using micro-coax, flat flex cables, or low-profile micro connectors instead ⁷¹. Thus KK connectors could be considered legacy in the context of modern mobile device design, though they thrive in legacy-compatible realms like PC hobby projects or industrial control panels.
- In **telecommunications and networking equipment**, older backplane connector systems (including some made by Molex) have been replaced by high-speed modular connectors. For example, legacy telecom systems might have used Molex VMEbus or CompactPCI connectors; today these are often replaced by advanced high-speed backplane connectors (some made by Molex, some by others like TE Connectivity). The legacy connectors persist only in existing infrastructure, whereas new designs adopt connectors that can handle multi-gigabit signals.
- In **automotive**, as discussed, the presence of legacy connectors is actually intentional for longevity. But even there, some formerly standard connectors become legacy when a new standard is agreed upon. A case in point is the OBD-II diagnostic connector – not made by Molex specifically, but as a standard, it succeeded various proprietary diagnostic connectors (some of which Molex did manufacture for certain car makers). Once OBD-II became mandated, the older ones became legacy and disappeared from new vehicles. Similarly, as vehicles move to **higher-voltage EV systems**, certain 12V connector styles might be considered legacy for high-power connections, being replaced by newly designed high-voltage interlock connectors.

Why connectors become legacy: In general, a connector becomes “legacy” when **either its technical limitations impede new requirements or a superior standardized alternative emerges**. In the case of the Molex 4-pin: it could not meet the evolving requirements for SATA drives (hot-swap, additional voltage rail) ⁵⁹, and a new standard (SATA) took over with broad industry support. Once widely adopted, the new connector rapidly displaced the old in its primary application. Another reason is **ergonomics and safety** – newer connectors often aim to reduce user errors and improve serviceability. The 4-pin Molex's tendency to

stick and the possibility of connecting it incorrectly (there is a chamfer key, but many users forced them in misaligned, damaging pins) made it a target for replacement by something more foolproof.

Yet, it's worth noting that "legacy" doesn't mean "vanished." Molex and other manufacturers continue to produce legacy connectors for the aftermarket and replacement parts. Many enthusiasts still use Molex connectors in custom projects (for example, to power multiple fans via a single plug, or when refurbishing vintage computers). In some niche applications, a legacy connector might even be preferred simply because it's **familiar, readily available, and time-tested**. As an Arrow Electronics article pointed out, mature connector products offer advantages like assured long-term availability and widespread distributor support ⁶⁵. Engineers sometimes deliberately choose an older connector for a new design if the cutting-edge performance of a new connector isn't needed; this can reduce cost and leverage existing assembly tooling ⁷² ⁵⁰. Thus, legacy Molex connectors continue to serve in secondary roles and legacy support scenarios, even as design-in for primary roles has moved on.

In conclusion of this section, Molex's legacy connectors, particularly the once-omnipresent 4-pin power plug, illustrate how technology evolution and standardization cycles work in the connector world. The "Molex connector" in the colloquial PC sense has, in a way, become a victim of its own success – so prevalent that it needed to be replaced by something more advanced when the time came. Molex as a company, however, has often been part of developing those next-generation connectors (like SATA and others), ensuring that even as specific products become legacy, the company's innovations continue to shape the state of the art.

Conclusion

From its origin as a maker of plastic molding compounds in the 1930s to its status today as a global leader in electronic interconnects, Molex's journey is a remarkable case of technological adaptation and influence. The development of Molex's connector technologies closely parallels the evolution of electronics over the past seventy years. In the post-war era, Molex's introduction of affordable pin-and-socket connectors helped enable the expansion of consumer appliances and the nascent electronics industry ¹⁴ ¹². By the late 20th century, Molex connectors had become virtually *universal* in personal computing, powering the world's PCs and peripherals ³. The technical overview in this paper highlighted how Molex's connector designs evolved – becoming smaller, more capable, and more diverse – to meet the changing needs of applications ranging from tiny wearables to heavy-duty automotive systems. Molex's portfolio today ranges from legacy **0.093" power plugs** to cutting-edge **0.35 mm pitch board-to-board connectors**, reflecting an ongoing commitment to innovation in connectivity.

The applications surveyed – robotics, consumer electronics, and automotive – demonstrate the breadth of scenarios where Molex connectors are employed. In each domain, certain Molex products became standard solutions (e.g., Mini-Fit Jr. in PC power supplies, sealed MX150 connectors in cars, modular KK connectors in appliances), underscoring how the company's focus on reliability and ease-of-use resonated with engineers. These connectors often form the unseen backbone of electronic systems, quietly ensuring power and signals flow correctly. When they perform flawlessly, end-users rarely notice them – which is arguably the highest compliment for a connector. At the same time, we have seen that even successful connectors eventually face obsolescence as new requirements arise. The 4-pin Molex PC connector's transition to legacy status in the SATA era is a prime example: it highlights how advancements (hot-swap, higher pin counts) drive new interface standards, and older ones gracefully retire from the spotlight ⁵⁹. Molex's strategy of balancing new product development with support for established connectors has allowed for

continuity in industries like automotive, where legacy connectors remain in use for many years by necessity

65 .

In conclusion, the history and applications of Molex connectors underscore a broader theme in engineering: the importance of standardization coupled with innovation. Molex's connectors became ubiquitous by solving practical connectivity problems in a standardized way – be it linking a power supply to a motherboard or an airbag sensor to a car's ECU. By establishing reliable standards, Molex enabled different manufacturers' components to connect seamlessly, which was crucial for the growth of complex electronic ecosystems. At the same time, the company did not stagnate; it continually developed new connector technologies (often in collaboration with industry consortia) to address emerging needs such as higher data rates, smaller form factors, and greater durability. This dual legacy of Molex – as both an innovator and a custodian of connector standards – has made its mark on virtually every area of electronics.

The legacy of Molex connectors is therefore twofold: **technological** (embodied in the connectors themselves, many of which became industry benchmarks) and **cultural** (the very word “Molex” entered the lexicon as a synonym for a certain type of connector). Even as specific Molex connectors become legacy, the company's influence persists in the connectors that replace them. Future trends like electric vehicles, 5G networks, and IoT devices will undoubtedly demand new connector solutions – and building on a foundation 80+ years strong, Molex is poised to continue playing a key role in “creating connections” for the next generation of innovation 73 .

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